

NAME: Jacob Golota

ChE 210

Quiz #1

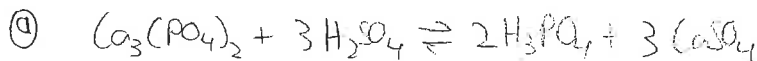
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To Receive Full Credit, You Must Show All Your Work.**Be sure to use units correctly in all of your calculations.**

Phosphoric acid (H_3PO_4) can be produced by reacting calcium phosphate ($\text{Ca}_3(\text{PO}_4)_2$) with a stoichiometric quantity of sulfuric acid (H_2SO_4). Calcium sulfate (CaSO_4) is also a byproduct of this reaction.

- (a) Write out the balanced chemical reaction equation. Make all of the stoichiometric coefficients the smallest whole numbers possible. Use any method you want to balance the reaction equation.
- (b) For 100 kg of H_2SO_4 , determine the kg of calcium phosphate consumed and the kg of phosphoric acid and calcium sulfate produced.
- (c) If calcium phosphate costs \$950/ metric-ton and sulfuric acid costs \$113/ metric-ton, and if phosphoric acid sells for \$960/ metric-ton and calcium sulfate sells for \$385/ metric-ton, what is the net profit or loss rate (\$/day) for the process based on selling 250 metric-ton/day of phosphoric acid produced only? Recalculate the profit/loss rate for this process considering that calcium sulfate can also be sold.
(1 metric-ton = 1000 kg)

Species	Molar Mass
$\text{Ca}_3(\text{PO}_4)_2$	310.18
H_2SO_4	98.08
H_3PO_4	97.994
CaSO_4	136.14



$$\textcircled{b} \quad 100 \text{ kg } \text{H}_2\text{SO}_4 \times \frac{1000 \text{ g}}{1 \text{ kg}} \times \frac{1 \text{ mol } \text{H}_2\text{SO}_4}{98.08 \text{ g } \text{H}_2\text{SO}_4} = 1019.58 \text{ mol } \text{H}_2\text{SO}_4$$

$$1019.58 \text{ mol } \text{H}_2\text{SO}_4 \times \frac{1 \text{ mol } \text{Ca}_3(\text{PO}_4)_2}{3 \text{ mol } \text{H}_2\text{SO}_4} \times \frac{310.18 \text{ g}}{1 \text{ mol } \text{Ca}_3(\text{PO}_4)_2} \times \frac{1 \text{ kg}}{1000 \text{ g}} = 105.42 \text{ kg } \text{Ca}_3(\text{PO}_4)_2$$

$$1019.58 \text{ mol } \text{H}_2\text{SO}_4 \times \frac{2 \text{ mol } \text{H}_3\text{PO}_4}{3 \text{ mol } \text{H}_2\text{SO}_4} \times \frac{97.994 \text{ g}}{1 \text{ mol } \text{H}_3\text{PO}_4} \times \frac{1 \text{ kg}}{1000 \text{ g}} = 66.61 \text{ kg } \text{H}_3\text{PO}_4$$

$$1019.58 \text{ mol } \text{H}_2\text{SO}_4 \times \frac{3 \text{ mol } \text{CaSO}_4}{3 \text{ mol } \text{H}_2\text{SO}_4} \times \frac{136.14 \text{ g}}{1 \text{ mol } \text{CaSO}_4} \times \frac{1 \text{ kg}}{1000 \text{ g}} = 138.8 \text{ kg } \text{CaSO}_4$$

Species	ν	M	$\nu \cdot M$	\$/ton	ton/day	\$/day
$\text{Ca}_3(\text{PO}_4)_2$	1	310.18	310.18	-950	395.66	-375877
H_2SO_4	3	98.08	294.24	-113	375.33	-427259
H_3PO_4	2	97.994	195.988	960	250	240000
CaSO_4	3	136.14	408.42	385	530.98	2005743

more work
on back



250 ton

$$\text{tot w/out } \text{CaSO}_4 = -178289.29 \$$$

$$\text{tot w/ } \text{CaSO}_4 = 22288.01 \$$$

$$250 \text{ ton } \text{H}_3\text{PO}_4 \times \frac{1000 \text{ kg}}{1 \text{ ton}} \times \frac{100 \text{ g}}{1 \text{ kg}} \times \frac{1 \text{ mol}}{97.99 \text{ g}} = 2551176.603 \text{ mol } \text{H}_3\text{PO}_4$$

$$2551176.603 \text{ mol } \text{H}_3\text{PO}_4 \times \frac{3 \text{ mol } \text{CaSO}_4}{2 \text{ mol } \text{H}_3\text{PO}_4} \times \frac{136.14 \text{ g}}{1 \text{ mol}} \times \frac{1 \text{ ton}}{1000000 \text{ g}} = 520.98 \text{ ton } \text{CaSO}_4$$

$$\times \frac{98.08 \text{ g}}{1 \text{ mol}} \times \frac{1 \text{ ton}}{1000000 \text{ g}} = 378.133 \text{ ton } \text{H}_2\text{SO}_4$$

$$\times \frac{310.18 \text{ g}}{1 \text{ mol}} \times \frac{1 \text{ ton}}{1000000 \text{ g}} = 395.166 \text{ ton } \text{Ca}_3(\text{PO}_4)_2$$

OK